Regression Discontinuity Designs

Foundations and Extensions

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RDD seminar

Based on the two manuals by
Cattaneo, Idrobo, and Titiunik (2020, 2024)

Seminar structure

1. **Foundations: theory**
   Sharp RD, plots, continuity methods, validation
2. **Foundations: application**
   Turkish municipalities and women's education
3. **Extensions: theory**
   Local randomization, fuzzy RD, discrete scores, multiple dimensions
4. **Extensions: applications**
   Senate elections, financial aid, probation, geographic boundaries

The theoretical topics follow the manuals' order. Each empirical block follows its part's theory.

The canonical sharp RD design

1. The sharp RD design and its local effect
2. RD plots
3. Continuity based estimation and inference
4. Validation and falsification

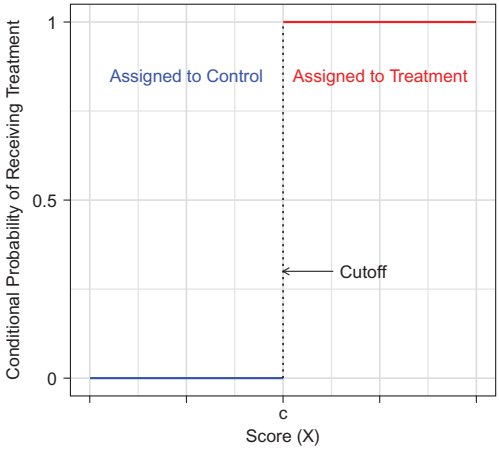
Foundations, Sections 1–5. (Cattaneo et al. 2020a)

The ingredients of a sharp RD design

- Each unit has a score X_i and faces a known cutoff c.
- The assignment rule is $T_i = \mathbf{1}\{X_i \geq c\}$.
- In a **sharp** design, actual treatment equals assignment: $D_i = T_i$.
- Foundations considers one continuous score, one cutoff, and perfect compliance.

Foundations, Section 2. (Cattaneo et al. 2020a)

Treatment assignment at the cutoff



Foundations, Section 2, Figure 1. (Cattaneo et al. 2020a)

The rule creates a discontinuous change in treatment probability.

Whether this change identifies a causal effect depends on assumptions about outcomes.

Potential outcomes and observed outcomes

For each unit, define $Y_i(1)$ and $Y_i(0)$.

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

- Only one potential outcome is observed for each unit.
- Above the cutoff we observe treated outcomes. Below it we observe untreated outcomes.
- Treatment effects require a comparison with the missing counterfactual.

Foundations, Section 2. (Cattaneo et al. 2020a)

Define the conditional mean potential outcomes

$$\mu_d(x) = \mathbb{E}[Y_i(d) \mid X_i = x], \qquad d \in \{0, 1\}.$$

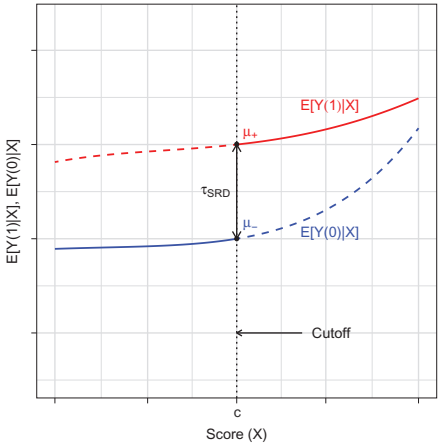
The target is the average treatment effect **at the cutoff**:

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = c] = \mu_1(c) - \mu_0(c).$$

- For a continuous score, conditioning at c describes a conditional mean function.
- The target does not require a large group of observations exactly at c.

Foundations, Section 2. (Cattaneo et al. 2020a)

Continuity of the counterfactual mean functions



Foundations, Section 2, Figure 2. (Cattaneo et al. 2020a)

Assume $\mu_0(x)$ and $\mu_1(x)$ are continuous at c.

The solid portions are observable. The dashed portions are counterfactual.

Continuity connects each observed limit to its missing value at the cutoff.

On the two sides of the threshold,

$$\mathbb{E}[Y_i \mid X_i = x] = \begin{cases} \mu_0(x), & x < c, \\ \mu_1(x), & x \geq c. \end{cases}$$

Continuity gives

$$\boxed{\tau_{\text{SRD}} = \lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x].}$$

- The observed outcome mean can jump because treatment changes.
- Another intervention changing at the same cutoff can undermine this interpretation.

Foundations, Section 2. (Cattaneo et al. 2020a)

The local nature of the effect

- The design identifies an effect for units at the assignment threshold.
- Treatment effects may vary with the score and other characteristics.
- A local effect can answer a relevant policy question about marginal eligibility.
- Effects far from c, or under a substantially different cutoff, require additional assumptions.

Identification and policy scope are separate questions.

Foundations, Section 2.2. (Cattaneo et al. 2020a)

What continuity allows

- The score can strongly predict both potential outcomes.
- Treated and control units can differ substantially away from the threshold.
- Continuity does not impose random assignment throughout a fixed interval.
- In a finite sample, estimation must approximate the limits using nearby observations.

Foundations, Sections 2 and 4.1. (Cattaneo et al. 2020a)

The purpose of an RD plot

- Show the relationship between the score and the outcome on each side.
- Reveal support, outliers, curvature, and a possible jump at c.
- Combine **local averages in bins** with separate smooth fits.
- Keep the graphical description distinct from formal estimation of the discontinuity.

Foundations, Section 3; Calonico et al. (2015). (Cattaneo et al. 2020a)

Partition the score into bins $I_1, \ldots, I_J$. For bin j,

$$\bar{X}_j = \frac{1}{n_j} \sum_{i:X_i \in I_j} X_i, \qquad \bar{Y}_j = \frac{1}{n_j} \sum_{i:X_i \in I_j} Y_i.$$

- Plot one point $(\bar{X}_j, \bar{Y}_j)$ per bin.
- Construct bins separately on each side so no bin mixes treatment statuses.
- A point summarizes many units. It is not generally an individual observation.

Foundations, Section 3. (Cattaneo et al. 2020a)

Evenly spaced and quantile spaced bins

Evenly spaced bins

Similar score width within each side.

Numbers of observations can vary considerably.

Sparse regions can produce noisy bin means.

Quantile spaced bins

Similar numbers of observations per bin.

Score widths vary with the density.

Dense regions receive a finer score partition.

Foundations, Section 3.1. (Cattaneo et al. 2020a)

The number of bins

- Few wide bins smooth the plot heavily and can hide local variation.
- Many narrow bins expose variation but make individual means less precise.
- Data driven bin selection formalizes the tradeoff.
- The two sides can use different numbers of bins.

Foundations, Section 3.2. (Cattaneo et al. 2020a)

IMSE and mimicking variance plots

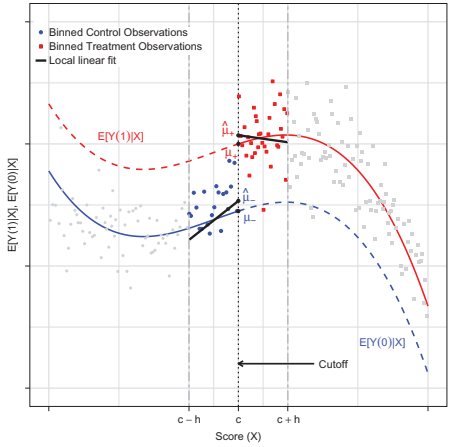
- **IMSE selection** balances integrated squared approximation error against variance when displaying the regression function.
- **Mimicking variance selection** uses more bins to reveal the outcome's variability.
- These are different graphical objectives and can produce different looking plots.
- A smooth fitted curve can accompany either choice.

Foundations, Sections 3.2.1–3.2.2. (Cattaneo et al. 2020a)

- Report the displayed score range and the binning rule.
- Fit curves separately on either side of the cutoff.
- A global polynomial used to describe a plot need not be the estimator used for causal inference.
- Confidence intervals for bin means do not provide a confidence interval for the jump.

Foundations, Sections 3 and 4.1. (Cattaneo et al. 2020a)

Local polynomial approximation



Approximate each regression function in a neighborhood of c.

Estimate its intercept at the cutoff.

Subtract the left intercept from the right intercept.

Foundations, Section 4.1, Figure 12. (Cattaneo et al. 2020a)

Let $x_i = X_i - c$ and $r_p(x) = (1, x, \ldots, x^p)'$. Separately for $s \in \{-, +\}$,

$$\widehat{\beta}_s = \arg\min_{\beta} \sum_{i \in \mathcal{I}_s} K\left(\frac{x_i}{h_s}\right) \left[Y_i - r_p(x_i)'\beta\right]^2,$$

where $\mathcal{I}_- = \{i : x_i < 0\}$ and $\mathcal{I}_+ = \{i : x_i \geq 0\}$.

$$\widehat{\tau} = e_1'(\widehat{\beta}_+ - \widehat{\beta}_-), \qquad e_1 = (1, 0, \ldots, 0)'.$$

Foundations, Section 4.2. (Cattaneo et al. 2020a)

With $p = 1$, the fitted lines are

$$\widehat{m}_-(x) = \widehat{\alpha}_- + \widehat{\beta}_- x, \qquad \widehat{m}_+(x) = \widehat{\alpha}_+ + \widehat{\beta}_+ x.$$

$$\widehat{\tau} = \widehat{\alpha}_+ - \widehat{\alpha}_-.$$

- The slopes can differ across sides.
- Centering the score at c makes each intercept a fitted cutoff value.
- Local quadratic regression adds curvature without imposing a global fit.

Foundations, Section 4.2. (Cattaneo et al. 2020a)

A single regression representation

Within a common window and using the same kernel weights,

$$Y_i = \alpha + \tau T_i + \beta x_i + \gamma T_i x_i + \varepsilon_i.$$

- Below the cutoff: intercept α, slope β.
- Above the cutoff: intercept $\alpha + \tau$, slope $\beta + \gamma$.
- The coefficient on T_i estimates the cutoff jump.
- The algebraic representation alone does not provide bias corrected inference.

Foundations, Section 4.2. (Cattaneo et al. 2020a)

Two common choices for the weights are

$$K_U(u) = \mathbf{1}\{|u| \leq 1\}, \qquad K_T(u) = (1 - |u|)\mathbf{1}\{|u| \leq 1\}.$$

- A uniform kernel gives equal weight within the chosen window.
- A triangular kernel gives more weight to observations near c.
- Kernel weighting localizes the regression. It does not randomize treatment.

Foundations, Section 4.2.1. (Cattaneo et al. 2020a)

The polynomial order

- Local linear regression is a practical default for estimating a level discontinuity.
- Local quadratic regression can check sensitivity to curvature.
- Higher orders can amplify variance and sensitivity near a boundary.
- An elaborate polynomial over the full score range can give distant observations substantial influence on the cutoff fit.

Foundations, Section 4.2.1. (Cattaneo et al. 2020a)

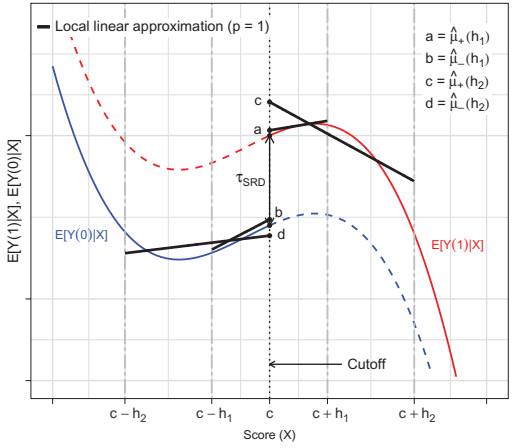
Bandwidth and local sample size

$$\mathcal{W}_h = [c - h_-, c + h_+].$$

- The bandwidth determines which observations inform the cutoff fit.
- Smaller h uses observations closer to c, reducing approximation error.
- Larger h generally improves precision by using more observations.
- The effective sample sizes on each side matter more than the overall dataset size for precision at c.

Foundations, Section 4.2.2. (Cattaneo et al. 2020a)

Approximation bias near the boundary



A line can approximate a smooth curve differently as the estimation window changes.

Foundations, Section 4.2.2, Figure 14. (Cattaneo et al. 2020a)

For a level discontinuity, under regularity conditions and a common h,

$$\text{Bias}(\hat{\tau}) \approx h^{p+1}\mathcal{B}, \qquad \text{Var}(\hat{\tau}) \approx \frac{\mathcal{V}}{nh}.$$

$$\text{MSE}(\hat{\tau}) \approx h^{2(p+1)}\mathcal{B}^2 + \frac{\mathcal{V}}{nh}.$$

- The squared bias and variance move in opposite directions.
- For $p = 1$, balancing their orders gives $h_{\text{MSE}} \propto n^{-1/5}$.

Foundations, Sections 4.2.2–4.2.3; corrected bias order in the 2023 erratum. (Cattaneo et al. 2020a)

Data driven bandwidth selection

- Feasible MSE selection estimates the unknown curvature and variance terms.
- A common bandwidth and separate bandwidths are both possible.
- The chosen bandwidth depends on the outcome, specification, and sample.
- Selecting a bandwidth to obtain a desired sign or significance level invalidates the stated analysis.

Foundations, Sections 4.2.2–4.2.4. (Cattaneo et al. 2020a)

Why conventional inference can fail

The conventional statistic treats approximation bias as negligible:

$$T_{\text{conv}} = \frac{\widehat{\tau} - \tau_0}{\widehat{\text{se}}(\widehat{\tau})}.$$

- At an MSE optimal bandwidth, bias is generally of the same order as the standard error.
- Centering a normal interval on $\widehat{\tau}$ can then produce poor coverage.
- Heteroskedasticity robust standard errors address sampling variability, but do not remove smoothing bias.

Foundations, Section 4.3. (Cattaneo et al. 2020a)

Undersmoothing and bias correction

- **Undersmoothing:** choose a smaller bandwidth so bias becomes negligible relative to sampling uncertainty.
- This sacrifices precision and requires a defensible rule for how much to shrink h.
- **Bias correction:** estimate the leading approximation bias and subtract it.
- Estimating that bias introduces additional uncertainty.

Foundations, Section 4.3. (Cattaneo et al. 2020a)

Estimate the bias using a higher order local polynomial, often $q = p + 1$:

$$\widehat{\tau}_{\text{bc}} = \widehat{\tau} - \widehat{B}.$$

Use a variance estimate that includes the uncertainty in $\widehat{B}$:

$$\text{CI}_{\text{RBC}} = \left[\widehat{\tau}_{\text{bc}} \ \pm \ z_{1-\alpha/2}\widehat{\text{se}}_{\text{RBC}}\right].$$

- Correcting the center without correcting the standard error is insufficient.
- The estimation bandwidth h and bias bandwidth b can differ.

Foundations, Section 4.3; Calonico et al. (2014). (Cattaneo et al. 2020a)

Point estimation and interval estimation

- **MSE optimal bandwidths** target the accuracy of the point estimator.
- **Coverage error optimal bandwidths** target the coverage accuracy of a confidence interval.
- Robust bias correction allows valid asymptotic inference at an MSE optimal bandwidth under appropriate conditions.
- A bandwidth optimized for one objective need not be optimal for the other.

Foundations, Section 4.3. (Cattaneo et al. 2020a)

Reading an RD results table

- The conventional point estimate is the difference between the fitted cutoff intercepts.
- The bias corrected estimate subtracts estimated smoothing bias.
- The robust interval centers on the bias corrected estimate and uses the corrected variance.
- A report should identify the estimator, interval method, polynomial order, kernel, bandwidths, and effective sample sizes.

Foundations, Section 4.3. (Cattaneo et al. 2020a)

Covariate adjustment

- Predetermined covariates can improve precision when they predict the outcome.
- A common additive adjustment can preserve the original RD effect under suitable balance and regularity conditions.
- Treatment affected variables can introduce post-treatment bias.
- Covariate adjustment cannot establish continuity when the underlying design is invalid.

Foundations, Section 4.4; Calonico et al. (2019). (Cattaneo et al. 2020a)

Dependence and effective information

- Standard errors must reflect the sampling and assignment environment.
- Clustering can be appropriate when outcomes share shocks within sampling groups.
- Few clusters near the cutoff can make conventional approximations unreliable.
- Repeated score values introduce a separate support problem, developed in Extensions.

Foundations, Section 4.4. (Cattaneo et al. 2020a)

What validation can establish

- The key continuity assumptions involve unobserved counterfactual outcomes.
- Observable implications can expose reasons those assumptions are implausible.
- Institutional knowledge is needed to interpret diagnostic evidence.
- Passing a collection of tests supports credibility, but cannot prove identification.

Foundations, Section 5. (Cattaneo et al. 2020a)

Predetermined covariates and placebo outcomes

- Use variables that the treatment could not have affected.
- Estimate their discontinuities with the same care given to the main outcome.
- Large or systematic jumps can indicate changes in composition or other confounders.
- With many covariates, isolated small p-values can arise by chance. Their magnitude and substantive meaning matter.

Foundations, Section 5.1. (Cattaneo et al. 2020a)

The density of the running variable

- A discontinuity in the score density can indicate precise sorting across the cutoff.
- Modern density tests compare one-sided density estimates directly.
- A density jump requires an explanation in the context of how the score is generated.
- A smooth density alone cannot exclude every form of sorting or confounding.

Foundations, Section 5.2; Cattaneo et al. (2020b). (Cattaneo et al. 2020a)

Placebo cutoffs

- Estimate artificial discontinuities at points where the treatment rule does not change.
- Restrict each analysis to observations on the same side of the actual cutoff.
- A placebo window crossing the true treatment threshold can pick up a real effect.
- Placebo jumps can reveal misspecification, other policies, or chance variation.

Foundations, Section 5.3. (Cattaneo et al. 2020a)

Observations closest to the cutoff

- Heaping or manipulation may concentrate at the threshold itself.
- A donut analysis removes observations very close to c.
- It then requires more extrapolation to recover a cutoff effect.
- Any change in results should be interpreted together with the reason for exclusion and the loss of local support.

Foundations, Section 5.4. (Cattaneo et al. 2020a)

Sensitivity to the bandwidth

- Compare a planned range of smaller and larger bandwidths.
- Examine magnitudes and confidence intervals, alongside sample sizes.
- Report how the bias bandwidth changes when h changes.
- Significance across several correlated specifications is not a sequence of independent confirmations.

Foundations, Section 5.5. (Cattaneo et al. 2020a)

A complete sharp RD analysis

1. Define the rule, outcome, and causal effect at the cutoff.
2. Inspect support and construct informative RD plots.
3. Estimate local polynomials with principled bandwidth selection.
4. Use inference that accounts for smoothing bias and dependence.
5. Assess predetermined outcomes, density, and specification sensitivity.
6. State the effect's local policy scope and remaining threats.

Foundations, Sections 2–6. (Cattaneo et al. 2020a)

Islamic political representation and women's education

The sharp RD illustration from Foundations, after the theoretical discussion.

Stata module 1 and worksheet A.

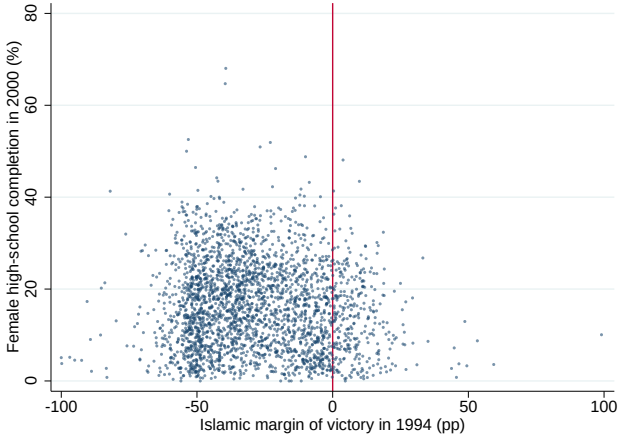
Original study: Meyersson (2014). Official companion code: CIT_2020_CUP_polecon.do.

The Turkish municipal election design

- **Units:** 2,629 municipalities in the authors' teaching dataset.
- **Score:** Islamic party margin of victory in the 1994 mayoral election, measured in percentage points.
- **Treatment:** an Islamic party wins the mayoralty, $T_i = \mathbf{1}\{X_i \geq 0\}$.
- **Outcome:** percentage of women aged 15–20 who completed high school, measured in 2000.
- **Target:** effect of an Islamic mayor for municipalities at a tied election.

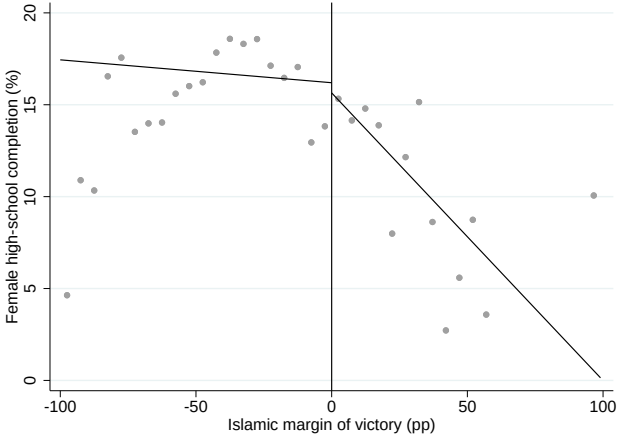
Meyersson (2014); Foundations, Section 2.1.

The raw outcome-score relationship



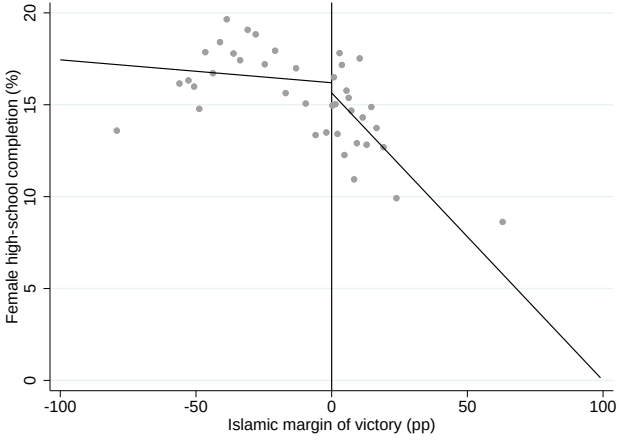
Stata module 1, authors' dataset. A comparison of all winners and losers mixes the cutoff effect with the outcome-score relationship.

Evenly spaced bins in the application



Twenty bins on each side, separate linear descriptive fits. Stata module 1; Foundations, Section 3.

Quantile spaced bins in the application



Twenty bins on each side, separate linear descriptive fits. Different bin widths reflect the uneven score density.

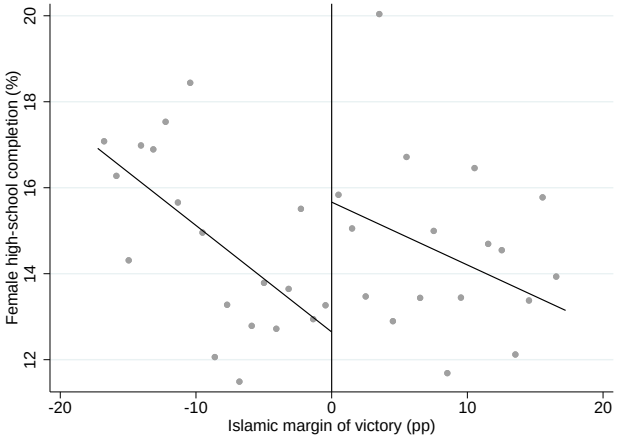
```
* An interaction allows different slopes on the two sides.
generate double TX = T*X
regress Y T X TX if abs(X)<=20, vce(robust)

* Data driven local estimation and robust inference.
rdrobust Y X, p(1) kernel(triangular) bwselect(mserd) all
rdrobust Y X, p(1) kernel(triangular) bwselect(cerrd) all
```

- Y is already a percentage. A coefficient of 3 means 3 percentage points.
- The robust RD interval accounts for estimated smoothing bias.

Foundations, Sections 4.2–4.3; module 1. (Cattaneo et al. 2020a)

The local fit around a tied election



Module 1. Local linear, triangular weights, MSE selected bandwidth. The plot uses the estimation sample and bandwidth.

Point estimates and robust intervals

	Estimate	Bias-corrected	Robust 95% CI
MSE bandwidth	3.02	2.98	$[-0.31, 6.28]$
CER bandwidth	2.43	2.41	$[-1.16, 5.98]$

- Units: percentage points of female high-school completion.
- The MSE point estimate is about +3.02 pp. Its robust interval includes zero.
- The data remain compatible with both a small negative effect and a substantively positive effect.

Saved Stata output, module 1. Current package results, rather than copied book values.

Predetermined covariates

	Estimate	Bias-corrected	Robust 95% CI
Log population, 1994	-0.08	-0.08	$[-0.81, 0.65]$
Number of parties, 1994	-0.26	-0.29	$[-1.58, 1.00]$
Islamic mayor, 1989	0.08	0.09	$[-0.07, 0.26]$

- Each row retains the covariate's own units.
- These checks use pre-treatment population, party count, and the previous mayor.
- Variables measured in 2000 can respond to the 1994 treatment. Their timing requires separate justification.

Module 1, coverage error selected bandwidths. A robust interval containing zero does not demonstrate exact balance.

Density evidence at the election threshold



Module 1, rddensity. Interpret the formal test together with election rules and opportunities for precise sorting.

Bandwidth and exclusion sensitivity

	Estimate	Bias-corrected	Robust 95% CI
Half bandwidth	1.80	1.26	$[-3.28, 5.81]$
Baseline	3.02	2.98	$[-0.31, 6.28]$
1.5 times bandwidth	2.97	3.19	$[0.32, 6.07]$
Local quadratic	2.77	2.66	$[-1.28, 6.60]$
Donut	3.75	3.85	$[0.41, 7.29]$

- The first three rows vary both h and b proportionally.
- The donut drops $|X| \leq 0.5$ and selects new bandwidths.
- A significant donut estimate does not override uncertainty in the full analysis.

Module 1. Outcome units are percentage points. Local quadratic uses its own selected bandwidth.

Sharp RD exercise and discussion

1. Explain why the full-sample mean difference is not the cutoff estimand.
2. Compare the two binning rules with the local fit.
3. Identify the point estimate and the center of the robust interval.
4. Evaluate the timing of each proposed balance covariate.
5. Write a conclusion that reports magnitude, uncertainty, and policy scope.

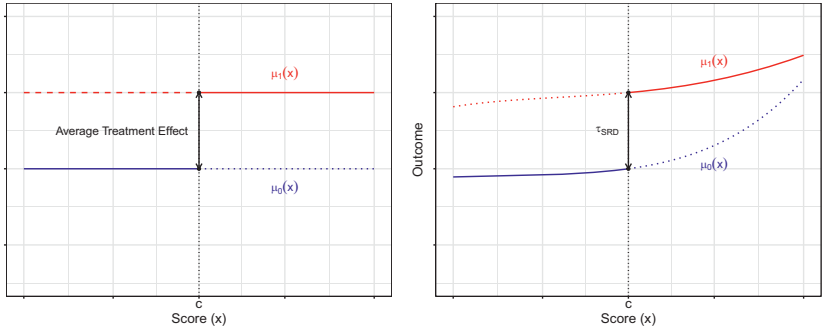
Worksheet A, module 1. Allow 35–45 minutes for running and interpreting the analysis.

Departures from the canonical sharp RD setup

1. Local randomization
2. Fuzzy RD
3. Discrete running variables
4. Multiple cutoffs and multiple scores

Extensions, Sections 1–5. (Cattaneo et al. 2024)

Continuity and local randomization



(a) Randomized experiment (b) RD design

- Continuity allows potential outcome means to vary with the score.
- Local randomization explicitly models assignment in a fixed window.

Extensions, Section 2, Figure 2. (Cattaneo et al. 2024)

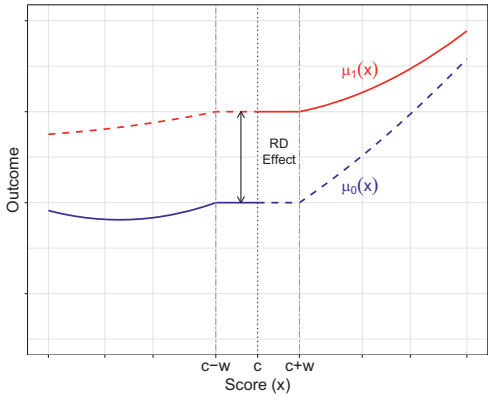
The local randomization assumptions

Choose a window $\mathcal{W} = [c - w, c + w]$ around the cutoff.

- **LR1:** within $\mathcal{W}$, the score is unconfounded with potential outcomes and the treatment assignment mechanism is known.
- **LR2:** within $\mathcal{W}$, changing the score has no direct effect on potential outcomes.
- Both assumptions need substantive justification in the chosen window.
- Randomness in the score alone does not rule out a direct score effect.

Extensions, Section 2, conditions LR1–LR2. (Cattaneo et al. 2024)

A different estimand inside the window



For fixed potential outcomes,

$$\theta_{\mathcal{W}} = \frac{1}{N_{\mathcal{W}}} \sum_{i:X_i \in \mathcal{W}} [Y_i(1) - Y_i(0)].$$

This is a window average.

The continuity target τ_{SRD} is an average at $X = c$.

Extensions, Sections 2 and 2.2.2, Figure 3. (Cattaneo et al. 2024)

The assignment mechanism

- **Complete randomization:** condition on the observed number treated in the window. All assignments with that count have equal probability.
- **Bernoulli assignment:** each unit receives treatment with a specified probability, allowing the treated count to vary.
- These mechanisms imply different randomization distributions.
- Choosing one is part of the research design and needs justification.

Extensions, Section 2.2.1. (Cattaneo et al. 2024)

The sharp null specifies every unit's treatment effect:

$$H_0^F : \; Y_i(1) = Y_i(0) \quad \text{for every } i \text{ in } \mathcal{W}.$$

1. Under this null, impute all missing potential outcomes.
2. Reassign treatment according to the stated mechanism.
3. Recalculate a statistic, such as the difference in means.
4. Compare the observed statistic with this randomization distribution.

Extensions, Section 2.2.1. (Cattaneo et al. 2024)

Exact tests and Monte Carlo approximations

- Enumerating every permissible assignment gives the finite assignment distribution under the sharp null.
- Simulation approximates that distribution when enumeration is costly.
- A fixed seed makes the simulation reproducible. More draws reduce Monte Carlo error.
- A displayed simulation p-value of zero means no simulated draw was as extreme. It is not evidence of a literally zero probability.

Extensions, Section 2.2.1. (Cattaneo et al. 2024)

Fisher intervals and average effect inference

- Invert sharp hypotheses $Y_i(1) - Y_i(0) = \theta$ to form a Fisherian interval under a constant effect model.
- A Neyman null instead concerns the average effect: $H_0^N : \theta_{\mathcal{W}} = 0$.
- Under complete randomization, the difference in means estimates the window average effect.
- Large sample inference for an average effect and Fisherian inference test different hypotheses.

Extensions, Sections 2.2.1–2.2.2. (Cattaneo et al. 2024)

Selecting the window

1. Specify a sequence of nested windows and predetermined covariates.
2. Evaluate covariate balance, beginning with the smallest window.
3. Enlarge the window while it meets the stated balance criterion.
4. Fix the rule before inspecting the main outcome results.

Failing to reject imbalance does not establish randomization. If the smallest window fails, the procedure may select no window.

Extensions, Section 2.2.4. (Cattaneo et al. 2024)

Validation under local randomization

- Examine predetermined covariates using the same window and assignment mechanism.
- Compare counts above and below the cutoff under a justified assignment probability.
- Use placebo cutoffs and sensitivity to nearby window choices.
- Selecting a window and then reusing balance evidence requires caution about what the tests establish.

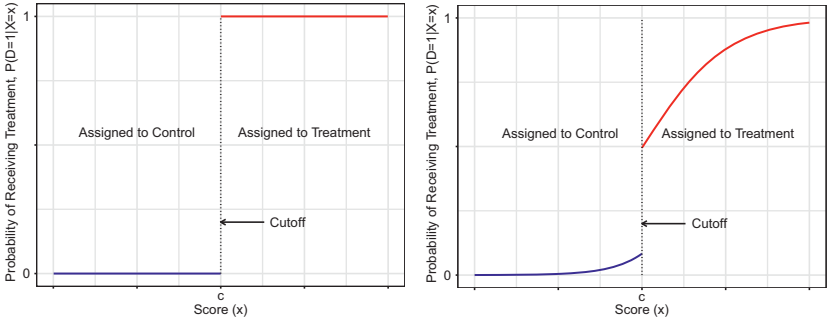
Extensions, Section 2.3. (Cattaneo et al. 2024)

When local randomization is useful

- It can support finite sample tests when a credible small window contains few units.
- With a continuous score, its fixed-window assumptions are typically stronger than continuity alone.
- With a coarse discrete score, it offers an alternative to extrapolation between unobserved score values.
- Different windows change both the units in the estimand and the plausibility of the assumptions.

Extensions, Section 2.4. (Cattaneo et al. 2024)

Assignment and receipt with imperfect compliance



(a) Sharp RD (b) Fuzzy RD

$$T_i = \mathbf{1}\{X_i \geq c\}, \qquad D_i \text{ is actual treatment received.}$$

The discontinuity in $\Pr(D_i = 1 \mid X_i = x)$ is nonzero and smaller than one.

Extensions, Section 3, Figure 7. (Cattaneo et al. 2024)

$$D_i = T_i D_i(1) + (1 - T_i) D_i(0).$$

- $D_i(t)$ describes receipt if assignment were set to t.
- $Y_i(t, d)$ allows assignment and receipt to affect outcomes separately.
- The observed outcome is $Y_i = Y_i(T_i, D_i(T_i))$.
- An exclusion restriction $Y_i(t, d) = Y_i(d)$ removes a direct effect of assignment on the outcome.

Extensions, Section 3. (Cattaneo et al. 2024)

Define the right-minus-left jump operator Δ_c.

$$\tau_Y = \Delta_c \mathbb{E}[Y_i \mid X_i = x], \qquad \tau_D = \Delta_c \mathbb{E}[D_i \mid X_i = x].$$

- τ_Y is the reduced form: the effect of assignment on the outcome under the relevant identification assumptions.
- τ_D is the first stage: the change in treatment receipt induced by crossing the cutoff.
- Assignment effects can be policy relevant even when the receipt effect needs further assumptions.

Extensions, Section 3.2.1. (Cattaneo et al. 2024)

Compliance types

Type	$D_i(0)$	$D_i(1)$
Never-taker	0	0
Complier	0	1
Always-taker	1	1
Defier	1	0

Monotonicity, $D_i(1) \geq D_i(0)$, excludes defiers. Compliance type is latent because only one potential treatment is observed.

Extensions, Section 3.2.2. (Cattaneo et al. 2024)

Under continuity of the relevant potential quantities, exclusion, a nonzero first stage, and monotonicity,

$$\boxed{\tau_{\text{FRD}} = \frac{\tau_Y}{\tau_D} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0), X_i = c].}$$

- This is a local average treatment effect for cutoff compliers.
- Always-takers and never-takers do not identify the receipt effect.
- A treatment effect for all recipients is a different parameter.

Extensions, Section 3.2.2. (Cattaneo et al. 2024)

Estimation of a fuzzy RD effect

$$\widehat{\tau}_{\text{FRD}} = \frac{\widehat{\tau}_Y}{\widehat{\tau}_D}.$$

- The reduced form and first stage must use compatible samples and local specifications for this ratio calculation.
- A joint fuzzy RD procedure selects bandwidths and accounts for covariance between numerator and denominator.
- Dividing two separately optimized point estimates need not reproduce a joint fuzzy estimate.
- Inference must account for smoothing bias in the ratio.

Extensions, Sections 3.2.2–3.2.3. (Cattaneo et al. 2024)

Fuzzy RD under local randomization

- Invoke the local randomization conditions within a credible window.
- Use assignment as an instrument for receipt within that window.
- Exclusion and monotonicity yield a window complier average effect, under appropriate assignment assumptions.
- Large sample TSLS inference and finite sample tests based on assignment require different arguments.

Extensions, Sections 3.2.1–3.2.3. (Cattaneo et al. 2024)

Weak assignment and validation

- A small or imprecisely estimated first stage makes the ratio unstable.
- Ordinary Wald intervals can fail under weak identification.
- Report the first stage and use inference suited to weak assignment when necessary.
- Covariate and density diagnostics concern the assignment rule. Exclusion and monotonicity need substantive arguments too.

Extensions, Sections 3.2.4–3.2.5. (Cattaneo et al. 2024)

A score with mass points

- Many units can share a score when the running variable takes discrete values.
- Count distinct score values overall and inside each estimation window.
- Inspect the closest score values on both sides and the treatment rule at exact ties.
- More observations at existing values improve precision there, but do not create observations closer to the cutoff.

Extensions, Sections 4.1–4.2. (Cattaneo et al. 2024)

The support gap around the cutoff

Let x_- and x_+ be the closest observed score values on each side.

$$\mathbb{E}[Y_i \mid X_i = x_+] - \mathbb{E}[Y_i \mid X_i = x_-]$$

is an observed cross-score contrast.

- If $x_- < c < x_+$ with a fixed gap, continuity alone does not identify the missing limits.
- A cutoff effect then requires restrictions on how outcomes behave between support points.
- A larger number of units at x_- and x_+ does not eliminate this problem.

Extensions, Section 4.3.1. (Cattaneo et al. 2024)

Continuity methods with a discrete score

- Local polynomial fits can be informative when there are many distinct values and the relevant smoothness assumptions are plausible.
- Inference still depends on how extrapolation between mass points is justified.
- Software adjustments for mass points improve practical bandwidth calculations.
- Clustering by the score does not itself resolve the identification or approximation problem.

Extensions, Section 4.3. (Cattaneo et al. 2024)

Local randomization with a discrete score

- Define a window containing a small, explicit set of score values.
- State why assignment across those values can be treated as randomized.
- Check predetermined covariates and assignment counts.
- The estimand becomes an effect for the units in that window, with assumptions that differ from smooth extrapolation to a missing cutoff.

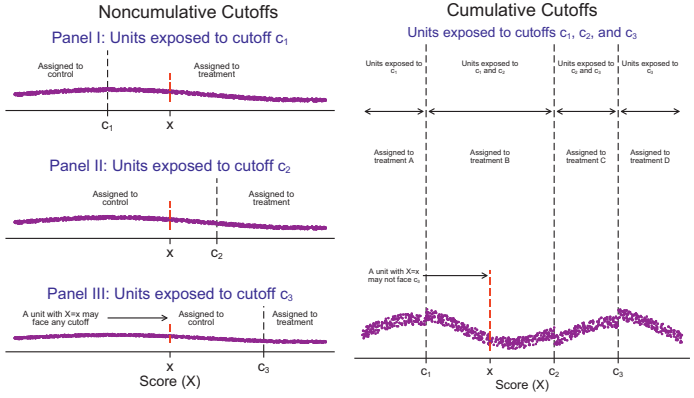
Extensions, Section 4.4. (Cattaneo et al. 2024)

Ties, selection, and interpretation

- Whether equality triggers treatment must agree with the actual rule.
- A tiny numerical shift can encode a tie rule, but it cannot supply new information.
- Missing outcomes may themselves result from treatment, such as outcomes observed only for people who remain enrolled.
- Identification of a score contrast does not automatically identify an effect for a stable population of observed survivors.

Extensions, Sections 4.1 and 4.4. (Cattaneo et al. 2024)

Multiple cutoffs: two assignment structures



(a) Noncumulative cutoffs (b) Cumulative cutoffs

Noncumulative: each unit faces one group-specific cutoff. Cumulative: the same unit faces several thresholds, often for different treatment levels.

Extensions, Section 5.1.1, Figure 12. (Cattaneo et al. 2024)

Cutoff-specific treatment effects

For a noncumulative design, let C_i denote the cutoff faced by unit i:

$$T_i = \mathbf{1}\{X_i \geq C_i\}, \qquad \tau(c) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = c, C_i = c].$$

- Each cutoff has its own local comparison and identifying assumptions.
- Differences between cutoff effects can reflect different populations and different score locations.
- Comparisons across cutoffs require attention to uncertainty and dependence.

Extensions, Section 5.1.2. (Cattaneo et al. 2024)

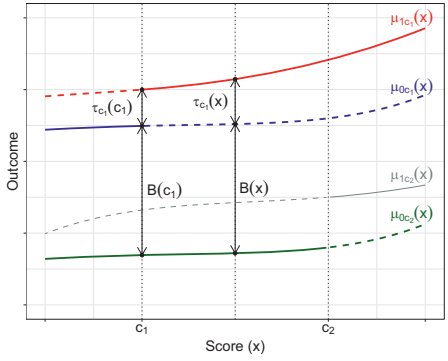
Define $\widetilde{X}_i = X_i - C_i$. All thresholds now occur at zero.

$$\tau_{\text{pool}} = \sum_{c \in \mathcal{C}} \omega(c)\tau(c).$$

- Under suitable conditions, the normalized discontinuity averages cutoff effects.
- The weights depend on cutoff-group frequencies and score densities at their thresholds.
- They generally differ from equal weights and full-sample group shares.
- Pooling obscures heterogeneity unless cutoff-specific effects are also shown.

Extensions, Section 5.1.2. (Cattaneo et al. 2024)

Extrapolation across cutoff groups



A constant untreated-outcome gap across score locations can link a lower-cutoff group's treated outcomes to a higher-cutoff group's controls.

Extensions, Section 5.1.2, Figure 14; Cattaneo et al. (2021). (Cattaneo et al. 2024)

The assumption behind extrapolation

For cutoff groups $c_L < c_H$, define

$$B(x) = \mu_{0,L}(x) - \mu_{0,H}(x).$$

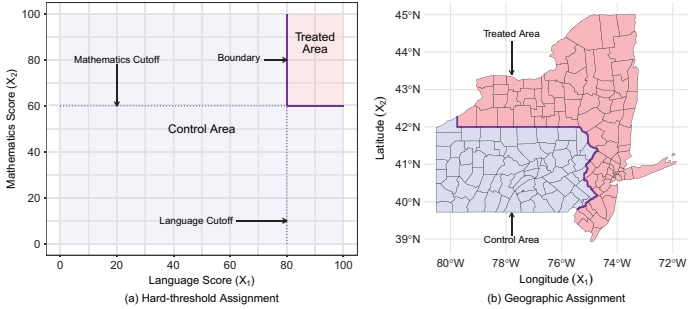
If $B(x) = B(c_L)$ for $x \in (c_L, c_H)$, then

$$\tau_L(x) = \mu_{1,L}(x) - \mu_{0,H}(x) - B(c_L).$$

- The lower-cutoff group is treated and the higher-cutoff group untreated at x.
- The constant-gap condition supplies the missing counterfactual. It is an additional identification assumption.

Extensions, Section 5.1.2. (Cattaneo et al. 2024)

Multiple scores and a treatment boundary



$$X_i = (X_{1i}, X_{2i}), \qquad T_i = a(X_i).$$

Treatment changes along a boundary $\mathcal{B}$, which contains many possible comparison points.

Extensions, Section 5.2, Figure 16. (Cattaneo et al. 2024)

Boundary point and pooled effects

For a boundary point $b \in \mathcal{B}$, the target is

$$\tau(b) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = b].$$

- A signed distance to a particular point can implement a point-specific comparison under appropriate conditions.
- Distance to the *closest point anywhere on the boundary* pools different locations.
- The pooled effect and point-specific effects answer different questions.
- Distance units and the metric affect which observations are close.

Extensions, Section 5.2.1. (Cattaneo et al. 2024)

Geographic boundaries

- Latitude and longitude jointly determine which side of a boundary a unit occupies.
- Geographic proximity can coexist with sharp differences in sorting, services, or institutions.
- Sparse settlement near the border increases extrapolation.
- Spatial dependence and spillovers can affect inference and interpretation.
- The causal claim requires a defensible treatment contrast at the chosen locations.

Extensions, Section 5.2.3. (Cattaneo et al. 2024)

Choosing the estimand and framework

Setting	Central question before estimation
Continuity, sharp RD	What effect do the outcome limits identify at c?
Local randomization	Which window and assignment mechanism are credible?
Fuzzy RD	What receipt effect is identified for which compliers?
Discrete score	What justifies the missing counterfactual across the support gap?
Multiple dimensions	Which cutoff group or boundary location defines the target?

Extensions, Sections 2–6. (Cattaneo et al. 2024)

Applications after the theory

1. Senate elections: local randomization
2. Ser Pilo Paga: imperfect compliance
3. Academic probation: a discrete score
4. Ser Pilo Paga and media markets: multiple dimensions

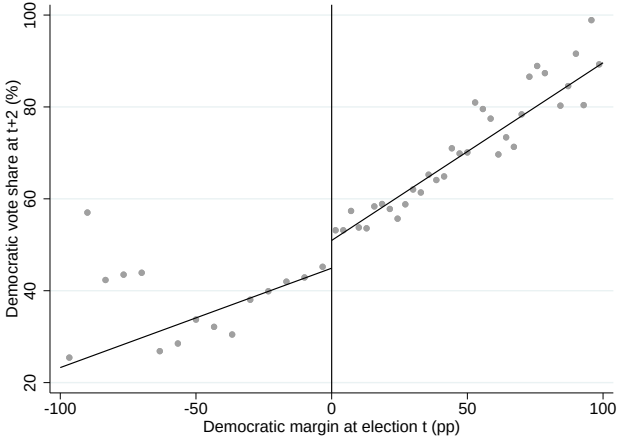
Stata modules 2–6 and worksheets B–F. The sequence follows Extensions.

Senate elections and party advantage

- **Units:** 1,390 U.S. Senate election observations in the teaching dataset.
- **Score:** Democratic margin of victory at election t.
- **Treatment:** a Democratic win at t.
- **Outcome:** Democratic vote share at election $t + 2$, the next election for that same seat.
- Here $t + 2$ indexes elections. It does not mean two calendar years later.

Cattaneo et al. (2015); Extensions, Section 2.1.

The Senate outcome-score relationship



Module 2. Separate linear descriptive fits and automatically chosen bins. Outcome and score use percentage units.

```
local covs presdemvoteshlag1 demvoteshlag1 demvoteshlag2 ///
    demwinprv1 demwinprv2 dmidterm dpresdem dopen
rdwinselect X ‘covs’, wobs(2) nwindows(50) approx seed(50)
local wl=r(w_left)
local wr=r(w_right)
rdrandinf Y X, wl(‘wl’) wr(‘wr’) reps(10000) seed(50)
```

- The selected window is approximately $[-0.7652, 0.7652]$ pp.
- Balance tests choose the window. The outcome does not enter that selection.

Extensions, Section 2.2; module 2. (Cattaneo et al. 2024)

Cutoff and window estimates in the Senate data

	Estimate	Bias-corrected	Robust 95% CI
Continuity at cutoff	7.41	7.51	[4.09, 10.92]

- The selected-window difference in means is 10.20 percentage points.
- None of 10,000 simulated null assignments gives a statistic as extreme as observed.
- The Fisherian test concerns the sharp null of zero effect for every unit in the window.
- The cutoff estimate and window estimate have different target populations and assumptions.

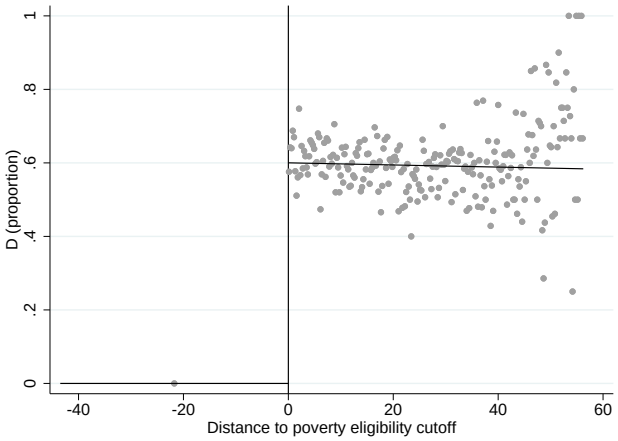
Module 2. Continuity table: percentage points, robust bias corrected interval.

Ser Pilo Paga and higher education access

- Colombia's programme combines academic merit and economic need.
- The one-score sample contains 23,132 students who meet the academic requirement.
- $X1$ measures poverty eligibility with its sign oriented so $X1 \geq 0$ means eligible.
- D records subsidy receipt. Y records immediate enrolment in any higher education institution.
- Eligible students need not take up the subsidy, so receipt follows a fuzzy design.

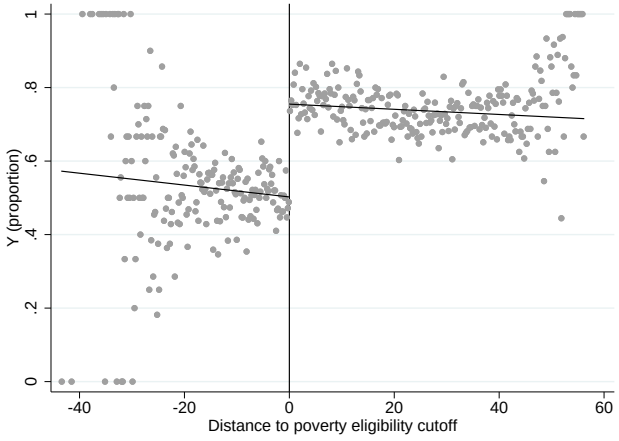
Londoño-Vélez et al. (2020); Extensions, Section 3.1.

The SPP first stage



Module 3. The vertical scale is the proportion receiving the subsidy, not a percentage.

The SPP reduced form



Module 3. The outcome is immediate higher education enrolment. The discontinuity estimates an eligibility effect.

Eligibility, take-up, and the receipt effect

	Estimate	Bias-corrected	Robust 95% CI
First stage (pp)	62.48	62.35	[59.52, 65.19]
Eligibility effect (pp)	26.90	27.41	[22.06, 32.75]
Receipt LATE (pp)	43.45	44.51	[36.60, 52.42]

- All effects and intervals in the table use percentage points.
- The fuzzy estimate is a receipt LATE for cutoff compliers under the stated assumptions.
- The first-stage row uses its own optimal bandwidth. Dividing the first two rows will not reproduce the joint fuzzy estimate.

Module 3. A common-window calculation in the do-file reproduces the conventional Wald ratio.

A window selection failure in the SPP data

- The module's approximate covariate balance procedure does not accept the smallest window at its stated threshold.
- It therefore selects no local randomization window.
- The code also shows the book's fixed $[-0.13, 0.13]$ window as an explicitly labelled illustration.
- That calculation does not establish a credible local randomization analysis under the failed selection protocol.

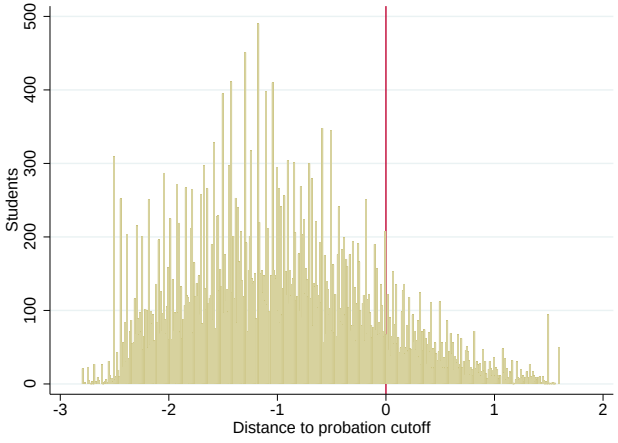
Module 3; Extensions, Sections 3.2–3.3. The teaching conclusion retains the failed diagnostic.

Academic probation and a discrete GPA score

- The teaching dataset contains 40,582 students and 429 distinct score values.
- The original rule assigns probation below a campus-specific GPA standard.
- The supplied transformed score is positive for probation and negative for control.
- Exact original ties are controls. The authors encode them as $X = -0.000005$.
- Next-term GPA is observed only for students who continue at university, creating a potential outcome-selection problem.

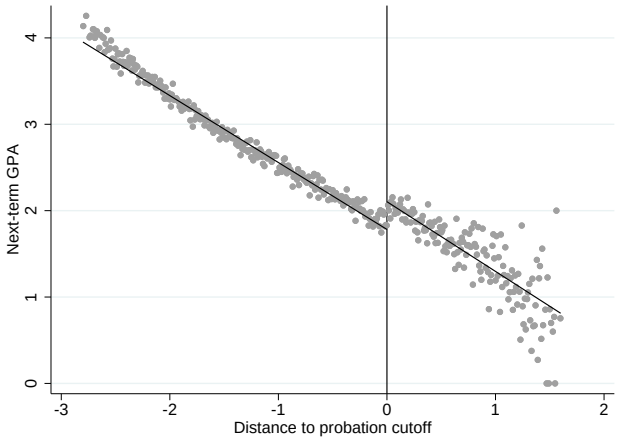
Lindo et al. (2010); Extensions, Sections 4.1–4.2.

The mass points in the probation score



Module 4. The large overall sample does not remove the discrete support. Closest groups: 208 controls at −0.000005, 67 treated at 0.01.

The probation outcome plot



Module 4. The score orientation makes lower prior performance correspond to a higher transformed score.

	Estimate	Bias-corrected	Robust 95% CI
Continuity method	0.22	0.22	[0.13, 0.30]
Cluster by score	0.22	0.21	[0.14, 0.28]

- Outcome units: GPA points. A point estimate near 0.22 is not 22 percentage points.
- In the closest two-score window, the difference is 0.234 GPA points and the simulated Fisherian p-value is about 0.057.
- Assignment counts and covariate diagnostics raise credibility questions. Clustering by score does not settle them.

Module 4. GPA among observed continuers does not automatically identify a causal effect for a common survivor population.

SPP with three area-specific cutoffs

- The merit-eligible sample faces different poverty thresholds by area.
- The supplied score reverses the original wealth-index sign.
- Metropolitan areas use $c = -57.21$, other urban areas $c = -56.32$, and rural areas $c = -40.75$.
- The outcome is enrolment, and the discontinuity is an eligibility effect.
- This dataset does not include receipt, so these estimates do not by themselves identify subsidy receipt LATEs.

Extensions, Section 5.1.3; module 5. (Cattaneo et al. 2024)

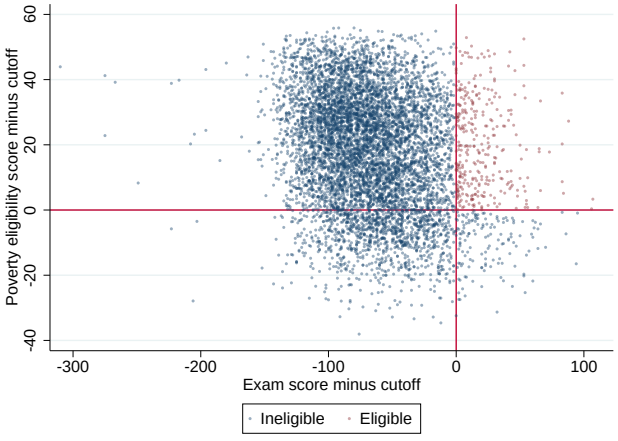
Cutoff-specific and pooled SPP effects

	Estimate	Bias-corrected	Robust 95% CI
Metropolitan (-57.21)	34.64	36.02	[26.86, 45.18]
Other urban (-56.32)	20.29	19.69	[11.16, 28.23]
Rural (-40.75)	20.89	18.27	[−4.24, 40.79]
Normalized pool	26.90	27.41	[22.06, 32.75]

- Units: percentage points of immediate enrolment.
- Rural estimates are substantially less precise.
- The normalized pooled estimate can conceal differences across groups.

Module 5. The do-file also reports rdmc's estimated aggregation weights.

SPP eligibility with both scores



Module 6, reproducible 2% plotting sample. Eligibility requires both centered scores to be nonnegative. Estimation uses the full supplied dataset.

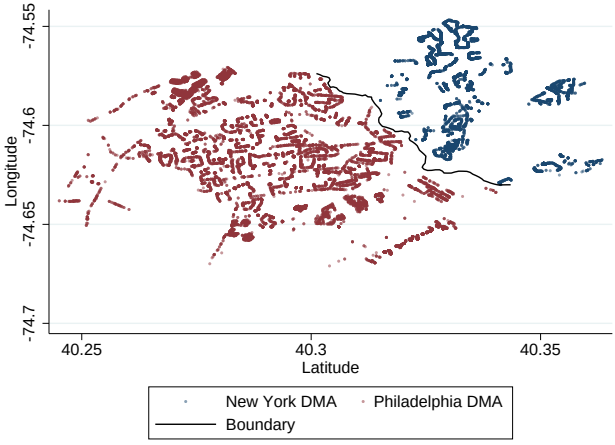
SPP effects at different boundary points

	Estimate	Bias-corrected	Robust 95% CI
(0, 0)	32.31	33.59	[29.30, 37.88]
(30, 0)	31.52	32.08	[28.58, 35.59]
(0, 50)	22.89	24.74	[14.39, 35.08]

- Coordinates are (poverty eligibility score, exam score), both centered.
- The data contain 574,269 student observations before outcome and score exclusions.
- Each row uses signed distance to its boundary point, in the authors' score units.
- These are eligibility effects in percentage points and need not coincide across locations.

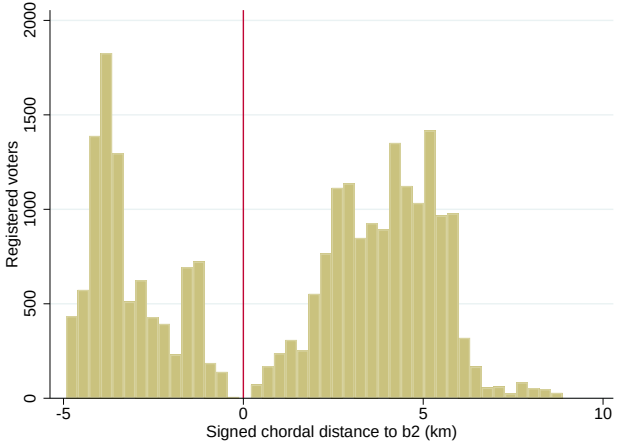
Extensions, Section 5.2.2; module 6. (Cattaneo et al. 2024)

Political advertising at a media-market boundary



Module 6; Keele and Titiunik (2015). Registered voters in New Jersey's West Windsor-Plainsboro school district. Outcome: turnout in the 2008 election.

Local support near the geographic boundary



Module 6. Signed chordal distance to the middle boundary point. The nearest treated and control locations remain roughly 0.31 and 0.42 km away.

	Estimate	Bias-corrected	Robust 95% CI
Boundary point 1	2.87	3.63	$[-6.40, 13.65]$
Boundary point 2	-0.20	-1.87	$[-19.74, 16.01]$
Boundary point 3	4.78	4.71	$[-19.64, 29.06]$
Whole-boundary distance	4.95	6.37	$[-8.34, 21.08]$

- Units: percentage points of turnout. Point-specific fits use signed chordal kilometres.
- The robust intervals are wide. They allow substantively meaningful effects in both directions.
- Distance to the whole boundary pools locations and changes the target.

Module 6; Extensions, Section 5.2.3. The reported baseline does not model spatial dependence.

Extensions exercises and discussion

- **B:** distinguish the Senate cutoff effect, window effect, and Fisherian null.
- **C:** explain SPP eligibility effects, receipt LATEs, and window selection failure.
- **D:** evaluate the probation tie rule, support gap, and outcome selection.
- **E–F:** interpret pooling, boundary-specific effects, distance metrics, and geographic uncertainty.

Worksheets B–F and Stata modules 2–6. Use selected modules in class and retain the others for a second lab.

References I

Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik (2019). "Regression Discontinuity Designs Using Covariates". In: *Review of Economics and Statistics* 101.3, pp. 442–451.

Calonico, S., M. D. Cattaneo, and R. Titiunik (2014). "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs". In: *Econometrica* 82.6, pp. 2295–2326.

— (2015). "Optimal Data-Driven Regression Discontinuity Plots". In: *Journal of the American Statistical Association* 110.512, pp. 1753–1769.

Cattaneo, M. D., B. R. Frandsen, and R. Titiunik (2015). "Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate". In: *Journal of Causal Inference* 3.1, pp. 1–24.

Cattaneo, M. D., N. Idrobo, and R. Titiunik (2020a). *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press.

— (2024). *A Practical Introduction to Regression Discontinuity Designs: Extensions*. Cambridge University Press.

Cattaneo, M. D., M. Jansson, and X. Ma (2020b). "Simple Local Polynomial Density Estimators". In: *Journal of the American Statistical Association* 115.531, pp. 1449–1455.

Cattaneo, M. D., L. Keele, R. Titiunik, and G. Vazquez-Bare (2021). "Extrapolating Treatment Effects in Multi-Cutoff Regression Discontinuity Designs". In: *Journal of the American Statistical Association* 116.536, pp. 1941–1952.

Keele, L. and R. Titiunik (2015). "Geographic Boundaries as Regression Discontinuities". In: *Political Analysis* 23.1, pp. 127–155.

References II

Lindo, J. M., N. J. Sanders, and P. Oreopoulos (2010). "Ability, Gender, and Performance Standards: Evidence from Academic Probation". In: *American Economic Journal: Applied Economics* 2.2, pp. 95–117.

Londoño-Vélez, J., C. Rodríguez, and F. Sánchez (2020). "Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia". In: *American Economic Journal: Economic Policy* 12.2, pp. 193–227.

Meyersson, E. (2014). "Islamic Rule and the Empowerment of the Poor and Pious". In: *Econometrica* 82.1, pp. 229–269.